

PROJECT_BRIEF_v1.5_FINAL_FREEZE.md

MonitorAI

Project Brief

Version 1.5 FINAL FREEZE

Status: Approved For Development

Project Name

MonitorAI

Project Type

iPhone-based Orthodontic Monitoring Platform

Project Goal

Develop an iPhone application capable of collecting highly standardized orthodontic intraoral photographs and metadata for:

- Remote monitoring
- Clinical review
- Future AI dataset generation

The application is not intended to provide diagnosis.

The application is intended to create a high-quality dataset and monitoring workflow.

Clinical Background

Orthodontic treatment increasingly relies on:

- Remote monitoring

- Reduced office visits
- Improved patient compliance

However, AI quality is limited by dataset quality.

The primary objective of MonitorAI is therefore:

Create a highly standardized orthodontic dataset.

Core Philosophy

MonitorAI prioritizes:

1. Standardization
2. Repeatability
3. Clinical usefulness
4. Metadata quality
5. Future AI value

over:

- Visual complexity
 - Development speed
 - Feature quantity
-

Target Users

Primary Users

Orthodontic Patients

Secondary Users

Orthodontists

Clinical Staff

Researchers

Target Devices

Supported:

iPhone 11 Series

iPhone 12 Series

iPhone 13 Series

iPhone 14 Series

iPhone 15 Series

iPhone 16 Series

iPhone 17 Series

Including:

- Standard
- Plus
- Pro
- Pro Max
- Air

Primary Development Device

iPhone 15 Pro Max

Technical Stack

Frontend

SwiftUI

Swift 6

MVVM

Backend

FileMaker Server 19

FileMaker Data API

Storage

External Secure Storage

Capture Workflow

Position Calibration



11-Step Capture Workflow



Review



Upload



Completion

Position Calibration

Before image capture:

The opening of the intraoral retractor must be aligned to screen center.

Requirements:

Center Error < 5%

Stable 3 Seconds

Automatic Advance

Capture Protocol

Total Images

11

Group A

Aligner ON

5 mm Open

Front

Right Buccal

Left Buccal

Group B

Aligner OFF

5 mm Open

Front

Right Buccal

Left Buccal

Upper Occlusal

Lower Occlusal

Group C

Light Occlusion

Front

Right Occlusion

Left Occlusion

Guidance System

Overlay PNG Guides

+

PTG

(Progressive Target Guidance)

PTG evaluates:

- Posterior visibility
- Occlusal visibility
- Clinical usefulness

PTG does not primarily evaluate head rotation angle.

Capture Logic

Automatic Capture

Rule:

$PTG \geq \text{Dot4}$

AND

Stable 4 Seconds

Plateau Capture

Rule:

$PTG \geq Dot4$

AND

No visibility improvement

AND

Stable 4 Seconds

Core Services

CameraService

PositionCalibrationService

OverlayGuideService

PTGService

VisibilityScoringService

AutoCaptureService

PlateauDetectionService

VoiceGuidanceService

HapticFeedbackService

MetadataService

StorageService

UploadService

Database Structure

Core Tables

Patients

MonitoringSessions

CaptureImages

CaptureMetadata

UploadQueue

TrainingLabels

SystemSettings

Metadata Strategy

Metadata is treated as equal in value to image data.

Every accepted image must store:

- PTG
 - Visibility
 - Focus
 - Blur
 - Exposure
 - Calibration
 - Timing
 - Device information
-

Dataset Integrity

Every image must additionally store:

ImageWidth

ImageHeight

ImageSizeBytes

ImageSHA256

Purpose:

- Duplicate detection
 - Integrity verification
 - AI export support
-

Offline Strategy

Capture workflow must operate without network access.

Uploads are queued locally and retried automatically.

Retry Schedule

1 minute

↓

5 minutes

↓

15 minutes

↓

1 hour

↓

Manual Retry

Security

Required

HTTPS

TLS 1.2+

Bearer Token

Keychain Storage

Dedicated API Account

External Secure Storage

Success Criteria

Patients can independently create clinically useful images.

Clinicians can review images reliably.

The resulting dataset supports future AI development.

Deliverables

iPhone Application

FileMaker Server Integration

FileMaker Data API Integration

Offline Queue System

Metadata Engine

AI-ready Dataset Structure

Final Statement

MonitorAI is not a camera application.

MonitorAI is a standardized orthodontic image acquisition platform designed to generate high-quality datasets for future orthodontic AI systems.